BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

LESSON 125 · Lesson 125 · Assembly Language and the Little Man Computer

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the purpose of a function return value?

Q2. After pushing A then B onto a stack, which is popped first?

Q3. What is the output of a full adder with A=1, B=1 and carry-in=0?

Q4. Why is the program counter important in a stored-program computer?

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

LMC — Educational model of a stored-program accumulator computer.

accumulator — Register holding current working data and arithmetic results.

program counter — Register holding address of the next instruction to fetch.

mnemonic — Symbolic name representing a machine instruction.

New learning

Mailboxes — Instructions and numeric data occupy addressed memory mailboxes. DAT reserves a data location. Programs execute the instructions addressed by the program counter.

Accumulator and PC — INP reads into ACC; LDA reads memory into ACC; STA writes ACC to memory. The PC normally advances after a fetch, unless a branch selects a new address.

Branches and stopping — ADD and SUB update ACC; OUT prints ACC; BRA always branches; BRZ tests for zero; BRP tests a non-negative result in standard LMC; HLT ends the program.

Worked example

Input is 7. Trace this program, including the accumulator, data mailbox and output.

1. Track each change — 00 INP: ACC = 7. — 01 STA 10: mailbox 10 becomes 7; ACC remains 7. — 02 LDA 10: ACC = 7. — 03 ADD 10: ACC = 14. — 04 OUT: output is 14 . — 05 HLT: stop. Data mailbox 10 is separate from instruction addresses 00–05.

We Do · Practise with guidance

Input is 9, and mailbox 21 contains 4. What is the output and why?

```
00 INP
01 STA 20
02 LDA 20
03 SUB 21
04 OUT
05 HLT
20 DAT 0
21 DAT 4
```

Predict the registers

What are ACC and mailbox 20 after the first two instructions?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

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Calculate ACC

Follow LDA 20 and SUB 21. Use the stored value 9 and constant 4.

Completed trace

INP sets ACC to 9. STA 20 saves 9. LDA 20 reloads it. SUB 21 subtracts 4, leaving ACC = 5. OUT prints 5 ; HLT stops.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Explain what the PC, accumulator and memory mailboxes represent in LMC. [6]

Question 2

Trace INP; STA 30; LDA 30; ADD 31; OUT; HLT, with input 5 and mailbox 31 containing 3. [6]

Question 3

Write LMC instructions to read an integer and output it, using a BRZ branch when zero. Explain how you would test both paths. [8]

Exit Ticket · Answer before checking

In LMC, STA n loads the accumulator with the contents of mailbox n.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It passes a computed result to the caller.

Q2. B, since a stack uses LIFO.

Q3. Sum 0, carry-out 1.

Q4. It tracks the address of the next instruction to fetch.

You Do pages · Independent questions

Q1. PC identifies next instruction to fetch [2]; accumulator holds data/arithmetic results [2]; mailboxes hold numbered instructions or data [2].

Q2. ACC=5 after INP [1]; mailbox 30 gets 5 [1]; LDA restores 5 [1]; ADD makes 8 [2]; OUT displays 8 [1].

Q3. One valid approach: INP; BRZ ZERO; OUT; HLT; ZERO OUT; HLT. BRZ branches if ACC=0 [2]; the nonzero path outputs entered number [2]; zero path outputs 0 [2]; HLT terminates and test values 0 and 7 exercise both paths [2].

Exit Ticket · Model answer

False. STA stores ACC into memory. LDA reads a value from memory into ACC.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: